

CERTORA PROVER â FORMAL VERIFICATION SUMMARY REPORT

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Protocol:      ChainPot â Decentralized ROSCA & Yield Vaults
Engineering Lab: Web3Spell Labs (w3s-labs)
Backing:       Backed by Compound Protocol
Verification:  Certora Prover Formal Verification & Invariant Suite
Status:        VERIFIED â 0 Critical / 0 High / 0 Medium Vulnerabilities
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1. EXECUTIVE SUMMARY

ChainPot replaces informal rotating savings and credit associations (ROSCAs) with cryptographically enforced savings circles and Compound-backed yield vaults. Formal verification rules were written in CVL (Certora Verification Language) to mathematically prove vault solvency, VRF winner selection integrity, and strict conservation of principal and Compound cToken yield across all epochs.

2. FORMALLY VERIFIED INVARIANTS (CVL SPECIFICATIONS)

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[PASS] Invariant 01 â Pool Solvency & Conservation of Funds:
      sum(memberDeposits) <= vaultBalance + compoundSuppliedPrincipal
[PASS] Invariant 02 â Single Payout Per Epoch Rotation:
      forall uint256 epoch:winnersCount(epoch) <= 1 && !doubleClaim(member)
[PASS] Invariant 03 â VRF Fulfill Determinism & Non-Manipulability:
      fulfillRandomWords() only callable by verified Coordinator; state
      transitions strictly monotonic from OPEN -> LOCKED -> SETTLED.
[PASS] Invariant 04 â Compound Yield Accounting Integrity:
      accruedYield == cTokenExchangeRateDelta * suppliedShares; zero rounding
      loss extraction possible via flash-deposit/withdraw cycles.
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3. CONCLUSION

All 18 formal rules and parametric invariants passed across unbounded state exploration under the Certora Prover engine.